

Table 1. *RBC-Nesbitt* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Nesbitt Hall	Philadelphia, PA, USA	Real	RBC	20	540	5-min	1 day

Table 2. Details of *RBC-Nesbitt* fault scenarios. Fault injection time varied in each case.

Fault abbreviation (CSV file name)	Fault type	Fault intensity	Fault injection date: yy/mm/dd (start-finish time)	Method of fault imposition
20160706_Chiller_Off	System stopped working (Chiller off)	-	16/07/06 (16:00-23:00)	Naturally occurred fault.
20160808_TSup_m4	AHU-1 supply air temperature bias	-4 °F	16/08/08 (10:22-21:16)	Overriding demand adjust by 2.
20160907_TSup_p4		+4 °F	16/09/07 (10:30-22:09)	
20160911_Chiller_Off	System stopped working (Chiller off)	-	16/09/11 (18:30-20:30)	Turning off the chiller from 6:30 p.m. to 8:30 p.m.
20161201_OADamStuck_90	AHU-2 outdoor air damper stuck higher than normal position	90% open	16/12/01 (10:00-20:45)	Overriding the AHU-2 outdoor air damper control signal* to be 90% open.
20170103_OA_DamStuck_80		80% open	17/01/03 (10:00-20:30)	Overriding the AHU-2 outdoor air damper control signal to be 80% open.
20170114_ScheduleOcc_early	Occupied during the unoccupied-scheduled period	-	17/01/14 (1:30-7:00)	Naturally occurred fault.
20170709_Chiller_Off	Chiller stopped earlier than scheduled	-	17/07/09 (4:00-15:30)	Naturally occurred fault.

20170711_OADamStuck_90	AHU-2 outdoor air damper stuck higher than normal position	90% open	17/07/11 (10:00-20:01)	Overriding the AHU-2 outdoor air damper control signal to be 90% open.
20170718_OADamStuck_100		100% open	17/07/18 (11:00-20:01)	Overriding the AHU-2 outdoor air damper control signal to be 100% open.
20170803_TCHWSup_m4	Chiller supply chilled water (CHWS) temperature sensor bias	-4°F	17/08/03 (10:00-21:27)	Overriding the CHWS temperature setpoint from 44 °F to 48 °F on control panel.
20170805_ScheduleOcc_early	HVAC system occupied earlier than scheduled	-	17/08/05 (4:00-8:00)	Changing the system schedule to be occupied from 4:00 a.m.
20170811_CooCoiValStuck_100	AHU-2 cooling coil valve position software override at higher than normal position	100% open	17/08/11 (10:05-20:06)	Overriding the cooling coil valve control signal to be 100% open.
20170915_DifPreCHW_p0.1	Chiller chilled water differential pressure (DP) sensor positive bias	0.1 psi	17/09/15 (10:30-15:21)	Overriding the DP setpoint from 7.5 to 0.5.
20180709_TSup_m3.5	AHU-2 supply air temperature sensor bias	-3.5 °F	18/07/09 (10:15-20:15)	Overriding the supply air temperature 'setpoint demand adjust' point by 1.75.
20180710_OADamStuck_30	AHU-2 outdoor air damper stuck at higher than the normal position	30% open	18/07/10 (10:30-20:30)	Overriding outdoor air damper control signal to be 30% open (normal mode: 15% open).

20180711_CooCoiValStuck_80	AHU-2 cooling coil valve stuck at higher than the normal position	80%	18/07/11 (10:00-20:00)	Overriding cooling coil control signal at 80% open in the BAS (normal mode: 40-60% open).
20180718_OADamStuck_60	AHU-2 outdoor air damper stuck at higher than the normal position	60% open	18/07/18 (9:30-19:30)	Overriding the outdoor air damper at 60% open in the BAS (normal mode: 15% open).
20180722_ScheduleOcc-wkend	Change weekend occupied schedule to end at 8:20 p.m.	-	18/07/22 (20:20-21:40)	Stopping the HVAC system at 8:20 p.m.
20180723_TCHWSup_m3	CHWS temperature sensor bias	-3.0 °F	18/07/23 (8:00-18:00)	Changing the CHWS temperature setpoint from 44 °F to 47 °F.

Additional information

- Extra helpful document(s):
 - Chen, Y. Data-driven Whole Building Fault Detection and Diagnosis. (Drexel University, 2019).
- This dataset contains 22 days of baseline data, representing behavior of the system when no artificial fault was injected. Please note, since this is a real system, the baseline and fault data do not share the same timestamps.
- Fig. 1. shows the schematic diagram of the AHU system used to generate the RBC-Nesbitt dataset, which reflects the actual system setup in the Nesbitt Hall.

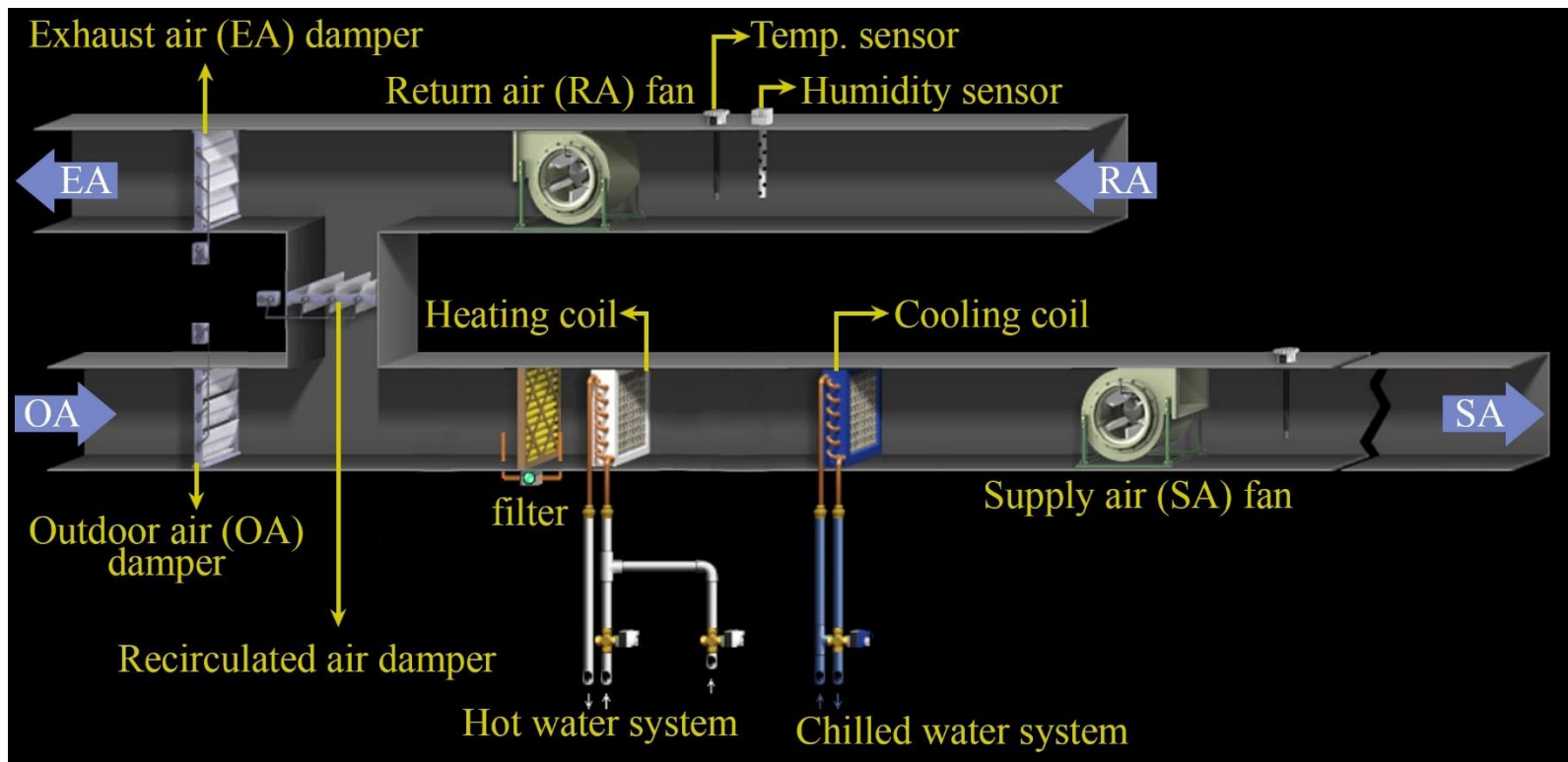


Fig. 1. The configuration of the AHU system, used to generate the *RBC-Nesbitt* dataset.